Faculty of Computers & Informatics
Academic Year: 2020-2021
First Year
Subject: Digital Logic
Number of Questions: 15

Second Term
Midterm Exam
Date: June 12, 2021
Allowed Time: Three Hours
No. of Pages: 2

Model (1)

(غير مسموح باستخدام الآلة الحاسبة فى الإمتحان)

(15x1=15 Mark)

Choose the correct answer

1. The decimal number, $(31)_{10}$ is equivalent to:
   (a) $(37)_8$ (b) $(43)_8$ (c) $(33)_8$ (d) $(30)_8$
2. The octal number, $(115)_8$ is equivalent to:
   (a) $(77)_{10}$ (b) $(98)_{10}$ (c) $(68)_{10}$ (d) $(74)_{10}$
3. The hexadecimal number, $(1F5)_{16}$ is equivalent to:
   (a) $(496)_{10}$ (b) $(512)_{10}$ (c) $(288)_{10}$ (d) $(501)_{10}$
4. Convert the binary number $(1011010)_2$ to hexadecimal:
   (a) $(5C)_{16}$ (b) $(5A)_{16}$ (c) $(5F)_{16}$ (d) $(5B)_{16}$
5. What is the two's complement of $(00001111)_2$?
   (a) 11110001 (b) 11110111 (c) 11110000 (d) 11111111
6. Solve this binary problem: 10000000 -00001111
   (a) 10000000 (b) 01111111 (c) 10000001 (d) 01110001
7. The number of bits used to store a BCD digit is:
   (a) 1 (b) 8 (c) 4 (d) 16
8. Based on the principles of Boolean algebra, the simplest form of $X = A(B + C) + C$ is:
   (a) $X = AB + AC$ (b) $X = AB + C$ (c) $X = A + B + C$ (d) $X = AB + AC + C$
9. By using DeMorgan's theorem, $X = \left(A'(B' + C)\right)'$ is simplified to:
   (a) $X = A(B + C)$ (b) $X = A + B + C'$
   (c) $X = ABC'$ (d) $X = A + BC'$
10. Which answer is an example of a product-of-sum (POS) expression?
    (a) $X = ABC$ (b) $X = (A + B)(C + D)$ (c) $X = AB + CD$ (d) $X = A + B + C$
11. The logic expression for NOR gates is:
    (a) $X = A' + B$ (b) $X = A + B'$ (c) $X = A + B$ (d) $X = (A + B)'$
12. The exclusive-OR (XOR), denoted by the symbol $\oplus$, is a logical operation that performs the following Boolean operation:
    (a) $x \oplus y = x y' + x' y$ (b) $x \oplus y = x y + x' y$
    (c) $x \oplus y = x y + x' y'$ (d) $x \oplus y = (x y') * (x' y)$
13. The exclusive- NOR is a logical operation that performs the following Boolean operation:
    (a) $x y' + x' y$ (b) $x y + x' y'$
    (c) $y + x y'$ (d) $(x y') * (x' y)$